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***The Impact of Dashboard Complexity on
Cognitive Load and Perceived Decision
Quality***

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Abstract

This study examined the impact of dashboard complexity on cognitive load and perceived decision quality in business dashboard environments. Drawing on Cognitive Load Theory, the research aimed to investigate whether visually complex dashboards increase users' mental effort and reduce confidence in decision making. Although business dashboards are widely used to support organizational decision making, previous research has provided limited empirical evidence regarding the psychological mechanisms through which dashboard design influences users' experiences and perceptions.

A quantitative repeated-measures survey design was adopted. Participants evaluated two dashboard conditions that differed in visual complexity and overall design quality. Dashboard A represented a clearer and more visually organized interface, while Dashboard B represented a more visually dense and cluttered interface. After viewing each dashboard, participants completed Likert-scale questionnaire items measuring dashboard complexity, cognitive load, and perceived decision quality. A total of 101 valid responses were included in the final analysis. Descriptive statistics and Pearson correlation analyses were conducted to examine relationships between the study variables.

The findings showed that the more visually complex dashboard generated substantially higher levels of perceived cognitive load and lower levels of perceived decision quality. Correlation analysis further demonstrated that higher dashboard complexity was positively associated with cognitive load, while higher cognitive load was negatively associated with perceived decision quality. The findings also suggested that cognitive load may partially explain the relationship between dashboard complexity and users' confidence in decision making.

This study contributes to business intelligence and data visualization research by providing empirical evidence that dashboard complexity can negatively influence users' cognitive experiences and perceptions of decision quality. The findings support Cognitive Load Theory and emphasize the importance of visual clarity, reduced clutter, and effective information presentation in dashboard design.

Keywords: business dashboards, dashboard complexity, cognitive load, perceived decision quality, business intelligence, data visualization

Résumé

Cette étude examine l'impact de la complexité des tableaux de bord sur la charge cognitive et la qualité perçue de la prise de décision dans les environnements de tableaux de bord d'entreprise. En s'appuyant sur la théorie de la charge cognitive, cette recherche visait à étudier si des tableaux de bord visuellement complexes augmentent l'effort mental des utilisateurs et réduisent leur confiance dans la prise de décision. Bien que les tableaux de bord soient largement utilisés pour soutenir la prise de décision organisationnelle, les recherches antérieures fournissent encore peu de preuves empiriques concernant les mécanismes psychologiques par lesquels la conception des tableaux de bord influence les perceptions et l'expérience des utilisateurs.

Une méthodologie quantitative à mesures répétées a été adoptée. Les participants ont évalué deux conditions de tableaux de bord présentant différents niveaux de complexité visuelle et de qualité de conception. Le tableau de bord A représentait une interface plus claire et mieux organisée visuellement, tandis que le tableau de bord B représentait une interface plus dense et encombrée visuellement. Après avoir observé chaque tableau de bord, les participants ont répondu à des questions sur échelle de Likert mesurant la complexité perçue du tableau de bord, la charge cognitive et la qualité perçue de la prise de décision. Au total, 101 réponses valides ont été incluses dans l'analyse finale. Des statistiques descriptives et des analyses de corrélation de Pearson ont été réalisées afin d'examiner les relations entre les variables étudiées.

Les résultats ont montré que le tableau de bord le plus complexe visuellement générait des niveaux plus élevés de charge cognitive perçue et des niveaux plus faibles de qualité perçue de la prise de décision. Les analyses de corrélation ont également démontré qu'une complexité plus élevée des tableaux de bord était positivement associée à la charge cognitive, tandis qu'une charge cognitive plus élevée était négativement associée à la qualité perçue de la prise de décision. Les résultats suggèrent également que la charge cognitive pourrait partiellement expliquer la relation entre la complexité des tableaux de bord et la confiance des utilisateurs dans leurs décisions.

Cette étude contribue à la recherche en business intelligence et en visualisation de données en apportant des preuves empiriques montrant que la complexité des tableaux de bord peut influencer négativement l'expérience cognitive des utilisateurs et leur perception de la qualité de leurs décisions. Les résultats soutiennent la théorie de la charge cognitive et soulignent

l'importance de la clarté visuelle, de la réduction de l'encombrement visuel et d'une présentation efficace de l'information dans la conception des tableaux de bord.

Mots-clés : tableaux de bord d'entreprise, complexité des tableaux de bord, charge cognitive, qualité perçue de la décision, business intelligence, visualisation de données

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1. Introduction

In recent decades, organizational data has begun to transform the way decisions are made in business settings. Advancements in business intelligence (BI) and data analysis are allowing organizations to process and visualize large amounts of data like never before (Burnay et al., 2024). Among these technologies are business dashboards which have emerged as a central tool for communicating insights (Yigitbasioglu & Velcu, 2012).

Business dashboards are visuals that display key performance indicators (KPIs), metrics, and trends to help with decision making (Few, 2016). By displaying data in a more concise view, dashboards reduce analytical effort and lead to faster decision-making (Nontokozo et al., 2025; Shao et al., 2024). As a result, dashboards have become prevalent across numerous industries, including finance, healthcare, education, manufacturing, and public administration (Nontokozo et al., 2025).

Although dashboards are becoming more common, evidence regarding the effectiveness of dashboards in improving users' decision making remains mixed. While some studies report positive effects on decision speed and productivity, others highlight challenges such as misunderstandings and poor decision outcomes when using dashboards (Eppler & Mengis, 2004; Bawden & Robinson, 2009; Burnay et al., 2024). These mixed findings suggest that the mere presence of dashboards does not guarantee better decisions and that additional factors shape how dashboards are used (Hjelle et al., 2024).

An assumption in early dashboard research was that more information led to better decisions, as long as the information is accurate (Brath & Peters, n.d.; Delone & Mclean, 2003). However, other research in psychology and information systems challenges this assumption by noting the cognitive limitations of the human brain during decision making. An example of this is the cognitive load theory, stating individuals possess limited attention and processing capacity, meaning an inability to process too much information at once (Sweller, 1988).

Additional research recognizes that dashboard effectiveness depends not only on what information is shown, but on how it is presented. Studies show that dashboards often suffer from visual clutter and poor information cues which can hinder users' ability to identify insights (Bera, 2016; Orlovskiy & Kopp, n.d.). In business contexts, dashboard characteristics such as information

format, visual design, and completeness influence users' comprehension and satisfaction, which in turn affect decision outcomes (Hjelle et al., 2024).

Much of the dashboard literature focuses on adoption and user satisfaction, often assuming that good dashboard usage equates to effectiveness and satisfaction (Burnay et al., 2024; Rieg, 2025). While focusing on user adoption in research offers insights into why dashboards are accepted or rejected, it is not clear how dashboards are perceived during actual use. As a result, the cognitive mechanisms linking dashboard design to confidence in understanding the information and decision making remain misunderstood (Hjelle et al., 2024).

Cognitive Load Theory offers a promising lense for addressing this. The theory shows that user's confidence in their decision making depends on cognitive resources and that information poorly displayed can impair confidence even when the information is accurate and relevant (Sweller et al., 2011). Applying this theory to dashboard research allows for a better understanding of why dashboards may fail to improve perceived decision confidence.

In response to this, the research problem that I will be addressing involves the limited understanding of how the design of business dashboards influences users' cognitive processing and confidence in their decision making that follows. While dashboards are widely adopted, existing research does not clearly explain how dashboard characteristics affect users' cognitive load and their evaluation of their confidence in decision making. The objective of this study is to examine the relationships between dashboard complexity, cognitive load, and perceived decision quality (how confident users were in their decision making). To address this objective, we will respond to the following research question: Does dashboard complexity increase cognitive load and reduce perceived decision quality? By focusing on cognitive load as a central factor, this research aims to contribute to a deeper psychological understanding of dashboard effectiveness in business contexts.

2. Literature Review

This literature review will examine how dashboard design characteristics have an impact on users' cognitive load. By synthesizing research from data visualization and cognitive psychology, the review aims to identify key gaps and to position cognitive load as a central mechanism in explaining dashboard effectiveness. This approach responds to recent calls for more theory-driven and cognitively grounded research on business dashboards and decision-making outcomes (Burnay et al., 2024; Hjelle et al., 2024).

2.1 Cognitive Load Theory

Cognitive Load Theory (CLT) can help us understand how people process information. Originally developed in the educational psychology field, CLT holds that human working memory has limited capacity and duration, so exceeding these limits results in impaired performance (Sweller, 1988; Sweller, Ayres, & Kalyuga, 2011). Although CLT started out as a way to understand learning, it is now being used to explain performance across multiple settings.

There are three types of cognitive load as defined by CLT: intrinsic, extraneous, and germane cognitive load (Sweller et al., 2011). Intrinsic cognitive load is the inherent complexity of a task, while germane load is the cognitive resources that are allocated to schema construction (these are generally seen as beneficial). On the other hand, extraneous cognitive load reflects the quality of presentation and interface design and is based on poor presentation and interface design (Sweller et al., 2011). Extraneous cognitive load will be of particular interest in the context of business dashboards, where visual design choices affect the amount of extraneous cognitive load, whereas the decision task itself is not responsible for the extraneous cognitive load.

Business dashboards are typically designed to provide a condensed representation of large amounts of data to facilitate quick and informed decision-making. However, if the information presented in dashboards lacks a rational or structured manner, there is a risk that users will experience cognitive overload. Recent dashboard research suggests that visual complexity and cluttered layouts on dashboards increase the cognitive effort needed to locate key information (Bera, 2016; Burnay et al., 2024). When an individual has to expend cognitive resources to just find their way through a dashboard rather than understand the data being presented, there are far fewer cognitive resources available for understanding.

Cognitive load is beginning to receive attention from researchers as it pertains to the use of dashboards. Burnay et al. (2024) presented an adoption model that identified three types of loads (informational, representational, and non-informational) and demonstrated that too much load had a negative effect on user's perception of the quality of the dashboard. Hjelle et al. (2024) investigated how formatting errors and incomplete information increased user's perception of task complexity, which is related to extraneous cognitive load. These studies provide evidence to support the notion that aspects of the design of dashboards can affect the quality of the user's decisions through cognitive processes, rather than just through the accuracy of the information presented.

However, even with these advancements in the understanding of cognitive load and dashboard use, it continues to be treated as a subordinate topic in the dashboard research literature. Many studies have attempted to indirectly measure the level of cognitive load (e.g., task completion times, error rates; Bera, 2016) rather than directly assessing users' perception of their level of mental effort. Therefore, cognitive load is largely discussed in the context of being an after-the-fact explanation rather than being considered a critical component of the overall discussion. As such, there is little evidence in the existing literature to show how aspects of dashboard design affect cognitive strain and how those strains will influence users' perception of the quality of their decisions. It will be necessary to reposition cognitive load, specifically extraneous cognitive load, as a primary variable that connects the complexity of a dashboard to users' confidence in their decisions.

2.2 Dashboard Design and Complexity

How users perceive, process and interpret information shown in dashboards are influenced by visual design decisions made by developers (Emami & Chau, 2020). Dashboards are usually judged based on the accuracy and usefulness of the data being displayed but the visual representation of the data affects how much cognitive energy will be needed during the decision-making process. Numerous studies have shown that visual representations can alter users' attention and the way they search for information (Few, 2016).

Dashboard design typically includes several types of visual representations (Rosenholtz et al., 2007), such as charts, tables, color coding, and interactive elements, and they are designed to

help users reduce the amount of analytical energy needed by making the cognitive processes more obvious by creating a visual representation of the data that makes it possible for users to see patterns. Alternatively, research demonstrates that visually coded data that is poorly designed requires a greater amount of cognitive energy than if the data was properly coded because users will have to visually search for the data and interpret it more than once (Bera, 2016). Therefore, visual design is not simply a method to deliver information; it is also a major determinant of the amount of cognitive energy required to process the information (Wu et al., 2022).

Visual complexity has been the main topic of research on dashboard design (Emami & Chau, 2020). This is the amount of visual elements that are presented in a single dashboard. For instance, if there are many charts, lots of color variations, or a lack of consistency, the dashboard will have visual complexity. It has been assumed that high levels of visual complexity provide a richer source of information; however, research demonstrates that increasing levels of visual complexity may negatively impact the amount of information users comprehend and how quickly they make decisions (Bawden & Robinson, 2009; Eppler & Mengis, 2004).

Color is a classic example of this principle. Color may help users to distinguish between values and provide attention to values; however, using too many colors can distract users and require more cognitive energy to process visual information (Wu et al., 2022). Excessive color variation has been shown to create longer decision times because users are attempting to figure out what color differences are relevant when in fact there are no meaningful differences in relation to the data (Bera, 2016). The findings align with the Cognitive Load Theory, which states that when there are visual cues that do not have a purpose, they create unnecessary cognitive load.

Those dashboards which do not exhibit a clear type of hierarchy force the user to shift their visual attention continuously between dashboard elements to find relevant indicators (Few, 2016). Furthermore, research by Hjelle et al. (2024) indicates there is often a tendency for users to rely on visually dominant elements in the dashboard regardless of whether they are relevant for completing a task. In addition to visual design, interactivity has a major impact on the complexity of a dashboard. Interactive dashboards allow users to filter, drill down into, and explore data which can contribute to deeper analysis; however, interaction can also increase cognitive demand since users may have to remember a number of interaction rules or views in order to complete a task.

Experts' varying levels of experience with visual designs can also be problematic. Visual designs that are efficient for experts' visual analysis may impose large amounts of cognitive load upon novices who lack existing schemas from which to interpret complex visualizations (Shao et al., 2024). As effort to using the dashboard increases, the cognitive load places on the novice user increases and subsequently the user's confidence in their decisions decreases. Despite this limitation, the majority of studies completed on dashboards use experienced or technically competent users; therefore, the implications of these types of studies on non-expert users are reasonably limited.

Ultimately, the research to this point indicates that cognitive effort is a major contributor to decision-making due to dashboard complexity. However, the majority of current research on this relationship has been inadequate theoretically and operationalized inconsistently. Although visual design has an effect on the cognitive processing of decision-makers, often the cognitive roots are inferred rather than evaluated. This evidence of a gap in research supports future exploration which both explicitly focuses on dashboard complexity as an extraneous cognitive load and tests the effect of dashboard design on decision quality by means of meditation; thus leading to future studies being grounded in cognitive load theory. By establishing a theoretical framework for analysing dashboard complexity, designs will lead more effectively to successful dashboard structures in business environments where novices and non-experts are increasingly being used to make operational decisions and may experience greater cognitive overload when using dashboards.

2.3 Cognitive Load and Decision Quality

Decision quality has become an important target for research investigating the use of business dashboards and decision support systems. Originally, dashboards are meant to display information; however, they are also intended as tools that will assist users with their decision-making quality. As of yet, the results of empirical tests of the effects of dashboards on decision quality have been mixed/found inconsistent. Cognitive load is increasingly being suggested as a potential linking mechanism between dashboard design and decision-making outcomes.

2.4 Conceptualizing Decision Quality in Dashboard Research

The definition of decision quality is the extent to which a decision is accurate or effective based on the information available. Decision quality has been measured in several ways in dashboard research, including objective accuracy and performance on tasks (Hjelle et al., 2024; Burnay et al., 2024).

Most dashboard studies assume that better access to information provides better quality decisions. Ultimately, this viewpoint fails to recognize the cognitive processes that must take place in order to interpret and combine the data from a dashboard. As a result, how well a decision quality is made is contingent not only upon whether or not the data is available but also whether or not the end user can process it properly.

Cognitive science research suggests that increasing the amount of data in itself will not result in improving upon decision quality. Furthermore, if the amount of data exceeds the user's ability to process that information, additional data could impair the overall quality of a decision (Eppler & Mengis, 2004; Bawden & Robinson, 2009). In the context of the dashboard, therefore, it follows that an extremely detailed or complex dashboard will likely not improve decision-making due to a significant cognitive burden being placed upon its users.

2.5 Cognitive Load as a Determinant of Decision Quality

Cognitive Load theory explains why Decision Quality can decrease in situations of high cognitive loading (having too many items of extraneous cognitive load taking up space in Working Memory means less Working Memory resources are left for Higher-Order Thinking processes such as Inference and Evaluation) (Sweller et al., 1988). Therefore, it is possible that Decision Makers will find it more difficult to integrate pieces of information into a coherent whole, resulting in Lower Decision Quality and Lower Confidence in their decisions.

Research on Judgment and Decision-Making supports the evidence of the relationship between Cognitive Load and Decision Making. Tversky and Kahneman proposed (1974) the Theory of Attention and Effort states that individuals allocate a limited amount of Cognitive Resources to competing tasks, and when the amount of competing demands exceeds the capacity to process them, Decision Making Performance will decline due to Cognitive Load.

In addition, under conditions of High Cognitive Load, individuals are more likely to use heuristics or simplify their Decision Strategies to make the decision quickly, possibly at the expense of Decision-Making Accuracy (Tversky & Kahneman, 1974). In a Dashboard Environment, this can be seen as an over-reliance on Visual Prominence of Metrics or failure to attend to less visually prominent but still relevant Information for Decision Making.

Recent empirical studies support the growing evidence that Cognitive Load plays a significant role in Decision Making supported by Dashboards. The study by Hjelle et al. (2024) demonstrates that higher levels of perceived task complexity have a negative effect on Decision Making Quality; whereas, higher levels of Information Satisfaction have a positive effect on Decision Making Outcomes. These findings suggest Dashboards impact Decision Quality not only through the quality of the information contained in the Dashboard but more so, by influencing the Users' Cognitive Experience.

Similarly, Burnay et al. (2024) found that Dashboards producing excessive Information Load have Lower Perceived Quality and Lower Effectiveness even when there are High Adoption Intentions. This study provides evidence for the broader literature on Information Overload, which consistently demonstrates that if there is excessive cognitive demand, individuals will be unable to accurately evaluate information and make appropriate decisions (Bawden & Robinson, 2009; Eppler & Mengis, 2004). Importantly, these negative consequences can occur even when the information is accurate and relevant, highlighting the distinct differences between Information Quality and Cognitive Efficiency.

2.6 Mediating Role of Cognitive Load in Dashboard Effectiveness

Despite interest in cognitive load, current dashboard studies still model decision-making quality as a direct result of aspects of the dashboard design (i.e., visual design). There are processes in a person's mind when they are viewing dashboards that can impact the decision i.e., cognitive processes. Therefore, with different studies producing mixed results with regard to dashboards having positive, negative, and little or no impacts on decision-making quality, the measurement of cognitive load as a mediating variable provides greater explanation and insight into how a dashboard design characteristic (i.e., complexity, poor information layout) can produce an

increased level of extraneous cognitive load when the person is using the dashboard, ultimately leading to decreased efficiency in processing information.

There is emerging evidence to support this, but it is limited. For example, Hjelle et al. (2024) provide non-direct evidence that suggests that perceived task complexity mediates the relationship between decision quality measures. However, the construct of perceived task complexity is a general term for multiple variables which may include cognitive load as well as variables that are not cognitive loads (i.e., task novelty, domain unfamiliarity). Burnay et al. (2024) provide evidence that cognitive load can influence the adoption of dashboards, but they do not show that cognitive load directly impacts decision-quality measures.

2.7 Synthesis of Research Gaps

An examination of the literature suggests that business dashboards are relatively common as a type of decision-support tool, but there has been little theoretical or empirical evidence to suggest that business dashboards improve the quality of business decisions. Previous literature has independently addressed the subjects of dashboard adoption, information quality, and visual design. Nevertheless, there remain significant knowledge gaps within the literature regarding how these three key subject areas interact.

Although cognitive load theory is widely cited within the context of dashboards and visualization studies, it is rarely applied as a central variable. Cognitive load is typically discussed for the purpose of justifying various design recommendations; however, few studies have actually explored cognitive load as a mediating mechanism that explains how certain dashboard characteristics affect decision outcomes (Sweller et al., 2011; Burnay et al., 2024). By failing to address cognitive load conceptually within the context of dashboards, it is impossible to explain why certain dashboards that contain high-quality information do not improve decision making when poorly designed.

Researchers studying dashboards often assume that the decision quality associated with the use of dashboards is self-evident, rather than measuring decision quality associated with the use of dashboards. For example, many researchers equate the use of a dashboard as evidence of improved decision quality, or use only objective performance measures to judge improved decision quality, yet many dashboards can also be judged using subjective measures such as perceived

decision quality, perceived confidence in a decision, and perceived trust in an analytical tool (Rieg, 2025). Subjective measures are especially critical in business environments because the degree to which decision makers are confident in their decision-making process impacts future use of the dashboard and overall reliance on analytical tools.

The gaps identified within the existing corporate dashboard research highlight the need for developing an integrated, theory-based approach to addressing the dashboard decision quality gap that models cognitive load as a mediating mechanism between dashboard design characteristics and decision quality. Such a model would assist in reconciling a number of inconsistent research findings and contribute to a more accurate understanding of the effectiveness of dashboards in business decision-making processes.

2.8 Conceptual Model

Building on cognitive load theory, this study proposes a conceptual model in which dashboard design characteristics influence decision quality indirectly through cognitive load.

In this model:

- Dashboard complexity represents the key independent variable that shapes users' cognitive experience during dashboard interaction.
- Cognitive load, specifically extraneous cognitive load, functions as a mediating variable capturing the mental effort imposed by dashboard design.
- Perceived decision quality is the dependent variable, reflecting users' subjective evaluation of the confidence and reliability of their decisions.

By positioning cognitive load as a mediator, the model moves beyond adoption-centric explanations and focuses on the cognitive mechanisms underlying dashboard-supported decision making. This approach aligns with prior calls for greater integration of cognitive theory in dashboard research and addresses limitations in existing empirical models (Burnay et al., 2024; Hjelle et al., 2024).

2.9 Hypotheses Development

2.9.1 Dashboard Complexity and Cognitive Load

Cognitive load theory suggests that extraneous cognitive load arises from suboptimal information presentation rather than from task complexity itself (Sweller et al., 2011). In dashboard environments, high concentration of visuals, excessive metrics, and poor visual hierarchy are likely to increase the mental effort required to locate and interpret information. Empirical studies consistently show that increased dashboard complexity is associated with longer task completion times, higher perceived difficulty, and reduced user satisfaction (Bera, 2016; Burnay et al., 2024).

Based on this theoretical and empirical evidence, it is expected that higher dashboard complexity will increase users' cognitive load.

H1: Higher levels of dashboard complexity are positively associated with higher levels of users' cognitive load.

2.9.2 Cognitive Load and Perceived Decision Quality

Cognitive load theory suggests that decision quality deteriorates when cognitive demands exceed individuals' processing capacity (Sweller et al., 1988). Under high cognitive load, decision makers are more likely to experience reduced confidence in their judgments (Tversky & Kahneman, 1974). Empirical dashboard studies provide support for this relationship, showing that increased cognitive or task complexity negatively affects decision-making quality and satisfaction (Hjelle et al., 2024; Burnay et al., 2024).

Therefore, higher cognitive load is expected to reduce perceived decision quality.

H2: Cognitive load is negatively associated with perceived decision quality.

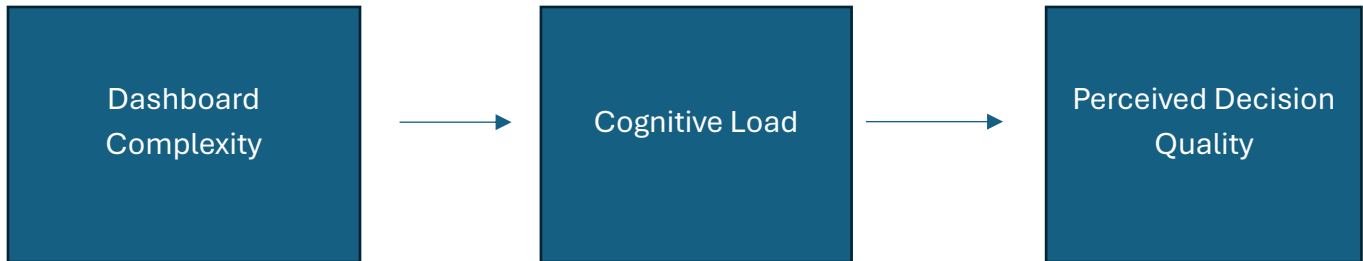
2.9.3 Mediating Role of Cognitive Load

Existing dashboard research increasingly suggests that dashboard design characteristics influence decision outcomes indirectly through cognitive mechanisms rather than directly (Hjelle et al., 2024; Burnay et al., 2024). However, few studies explicitly test cognitive load as a mediating variable between dashboard complexity, information quality, and decision quality.

By integrating cognitive load theory into the analytical model, my study proposes that dashboard complexity and information quality affect perceived decision quality through their impact on cognitive load.

H3: Cognitive load mediates the relationship between dashboard complexity and perceived decision quality.

3. Methodology



3.1 Research Design

This study used a quantitative research design to examine how dashboard complexity influenced users' cognitive load and their perceived decision quality. A quantitative approach was the best fit because the objective of the study was to measure relationships between clearly defined variables using numerical data collected through a structured questionnaire.

As discussed in the literature review, Cognitive Load Theory suggests that individuals have limited working memory capacity and that poorly structured information presentation can increase extraneous cognitive load (Sweller, 1988; Sweller et al., 2011). In dashboard environments, visual complexity and cluttered layouts may require additional mental effort to identify relevant information. This can negatively affect the confidence users have in their decisions and as a result, the performance and management of an organization.

To investigate the relationships between the variables, the study used a repeated-measures survey-based experimental design in which participants evaluated two different dashboard interfaces. The dashboards represented two levels of dashboard quality and visual complexity: Dashboard A represented a higher-quality dashboard designed according to commonly recommended data visualization principles, with clearer organization, balanced color usage, and reduced visual clutter, while Dashboard B represented a lower-quality dashboard containing a greater number of visual elements, excessive color usage, and a more visually dense layout (Bera, 2016; Few, 2016; Rosenholtz et al., 2007). The two dashboard types also aimed to simulate different levels of dashboard complexity that users may encounter in real business contexts.

Each participant was asked to review images of the two dashboards and to assume they were managers reviewing business performance data. Both dashboards contained an identical type of business information for the purpose of continuity of information. Following the review of each dashboard, respondents were asked to complete a survey measuring their perception of the degree of visual complexity and cognitive load experienced while reviewing the dashboard as well as their level of confidence in the decision they would make based on information from the dashboard. The survey contained Likert-type items to evaluate respondents' perceptions of visual complexity, mental effort associated with using the dashboard, and confidence in decision-making; the same set of items was used for both of the dashboard types to make comparisons possible.

I surveyed to determine how different designs affect users' subjective experiences while using visualized business data in dashboards. I was able to assess the relative influence of two types of dashboard conditions by comparing participant responses for both types of dashboards as well as assess whether there was a correlation between:

- 1) A higher amount of visual complexity on a dashboard produces greater cognitive load,
- 2) A greater cognitive load decreases participants' perception of the quality of their decisions (i.e., through perceived quality).

The study design closely reflected the conceptual model in the literature review; with dashboard complexity acting as an independent variable, cognitive load as a mediating variable and perceived decision quality as a dependent variable. I was able to provide empirical support for these theoretical constructs measuring them through structured survey items.

Given that I intended to solicit responses from a relatively large sample size, I chose a quantitative survey approach for data collection. The quantitative data collected and analyzed will allow me to identify patterns among user perceptions and conduct statistical data analysis of the relationships posited in the hypotheses. This methodological approach is common in business intelligence research when investigating user perceptions of technology and decision-support tools.

The research design allowed me to specifically answer the research question: Does increasing the complexity of a dashboard result in increased cognitive loads and therefore lower the perceived quality of the decisions made by users? By conducting a controlled comparison of the dashboards and collecting survey measures of the variables, I obtained practical methodological support to evaluate the theoretical links between the constructs in the literature review.

3.2 Participants and Sampling

Subjects for the study included undergraduate and early-career business professionals who were somewhat familiar with business-related information or data visualization. Business professionals were chosen for this study as the majority of organizations today utilize dashboards as an avenue for communicating important KPIs and assisting managers with making informed and effective decisions. As such, the individuals in this study would be representative of individuals who have an acceptable level of experience and exposure to business-related data (KPIs) through the use of analytical tools, and should afford an opportunity to evaluate how dashboard design impacts cognitive processing and the perception of decision quality as it relates to the initial premise of the research study.

Prior studies in the fields of business intelligence (BI) and data visualization commonly utilized participants with varying levels of analytics experience among students and junior professionals. This is due to the fact that the bulk of the dashboards users represent a dynamic market segment of organizational users today, and contain a broad spectrum of user profiles. Historically, dashboards have primarily been utilized by technical analysts, however, dashboards are now widely utilized by business professionals who depend on visual representation of data to assist them in making operational and strategic decisions. Examination of the cognitive processing and interpretation of dashboards by those with moderate levels of analytical experience will lend support to issues of insightfulness and relevance in the business world today.

The recruitment process for participants was achieved through convenience sampling methodologies. However, in retrospect, recruiting participants through the use of convenience sampling posed some unexpected limitations. First, the initial recruitment process through LinkedIn was ineffective, as only a few responses were generated, and as technical analysts and

other individuals who have some familiarity with business data represent a growing segment of the overall population, therefore, the recruitment of participants would require a greater effort. However, as the survey was distributed through the EDC Paris Business School's student and alumni networks, the number of participants responded to the survey increased substantially. This may be attributable to the likelihood of these networks possessing participants who are more experienced with business dashboards and utilizing data visualization tools than other participants. The result will ultimately influence and shape the final positions of the sample in the recruitment process.

The final dataset used 101 valid responses. The sample size was below the target of 150 responses, however was still large enough for exploratory analysis through descriptive statistics and correlation analysis. Using a large number of responses increased the reliability of the findings due to the reduced effects of individual variance on results.

To ensure the data collected were of high-quality, any responses that were incomplete or demonstrated patterns of inattentive response behaviour were removed from the final dataset. Valid responding was only measured in the statistical analysis based on completion of the items on the survey.

The sampling strategy for this research was designed to create an exploratory sample for the purpose of identifying general patterns in how dashboard users perceive their experiences. The goal of the research is to understand how dashboard complexity, cognitive load and perceived quality of decisions are related.

The study was conducted on a voluntary basis and all participants received information about the purpose of the research prior to completing the survey questionnaire along with explicit consent. Responses to the survey included assurances of anonymity and that the responses would be used for academic purposes only. All participants were required to complete this section in order for their responses to be valid.

3.3 Research Instrument

In this way the main data collection tool that was used in this research project was an online questionnaire to assess people's views about dashboard Complexity, cognitive load, and the quality of the decisions made based on the dashboards used by businesses. In constructing the Survey

items I created instruments based on the work of Burnay et al. (2024) which validated the scales they used to assess representativeness of the dashboards, Rieg (2025) who validated the scale they used to assess the beliefs of decision makers regarding the quality of their decisions as well as Wixom and Todd's (2005) Information Quality Framework, which Hjelle et al. (2024) did apply to test how well the dashboard they created measured the quality based upon format and completeness of the information provided.

The online questionnaire was conducted using an online survey company and included a number of sections which addressed the specific variables examined in the research model. The participants were exposed to screenshots of two business dashboards that displayed performance related metrics.

These performance measure dashboards contained the same types of business metrics and visualizations however they were different in both their visual layout and the level of Complexity associated with them. Specifically, Dashboard A represented a more organized and visually simple interface with a more defined structure, reduced visual clutter but also had a clearer path of flow throughout, where as Dashboard B had a more complexly structured interface that contained a greater number of visual elements contained in the dashboard presentation that would contribute towards the dashboard complexity and dashboard quality (as discussed previously).

Once the participants had viewed either screenshot of the business performance dashboards they were then required to imagine that they were representing a manager who was reviewing the business performance of their business and after each bridge/dashboard screenshot had been shown the participants completed a series of questions relating to their experiences with interpreting the presented dashboard. By utilizing this procedure, the research will gain feedback from the participants on their perceptions of dashboards immediately following their interactions with each of these dashboard stimulation's.

All of the questionnaire measurement items were expressed in a Likert-type format where respondents provided data on their perceptions of the degree of agreement they had with a series of statements utilizing a five (5) point response scale in which respondents indicated 'strong disagreement' > 'strongly disagree', 'disagree', neutral, 'agree', or 'strongly agree'. A Likert scale has been widely utilized in the field of information systems and social behavioral science to provide a consistent way to assess people's perceptions/attitudes/cognitive experiences with

respect to the subject being chosen for study. In this particular study a Likert scale was utilized so that the participants' response would reflect the amount of cognitive effort that they used in the process of interpreting the dashboards and also the amount of confidence the participant when making decisions based upon the data they had received through each dashboard.

Three core constructs were measured for this study, with participants responding to questions about their perceptions of dashboard complexity in terms of visual complexity or clutter (1st construct), their cognitive load in interpreting data on the dashboard (2nd construct), and decision quality confidence (3rd construct).

Each construct measured with 4 or more Likert scale items for thoroughness. The single reverse coded cognitive load item ("The dashboard was easy to interpret") served to help improve response consistency and reduce acquiescence bias.

The study used statements in questionnaire items to represent each of these constructs. Examples of these statements include "The clarity of this dashboard was easy to read", "The cognitive load exerted by interpreting this data was high", and "I could make confident decisions based on the data from this dashboard". Participants rated their level of agreement with each statement using a Likert scale.

Each participant completed a separate set of questionnaire items after viewing each dashboard. Participants' responses to the same set of questionnaire items completed after each dashboard allowed me to compare participants' perceptions of these dashboards, as well as analyze how visual complexity differences impacted cognitive load and perceived decision quality.

Collecting participant responses immediately after they viewed each dashboard allowed me to capture participants' subjective experiences engaging with these data visualizations. Therefore, I was able to examine the associations between dashboard complexity, cognitive load, and perceived decision quality as proposed by the research model.

3.4 Variables and Measurement

The purpose of this research paper was to explore the connections among dashboard complexity, cognitive load and the way people view how good decisions can be made. Each of these three variables that make up the overall model developed in this paper is defined

(independent variable – dashboard complexity, mediating variable – cognitive load and dependent variable - perceived decision quality) by an operational explanation of how to measure them.

The variable that was manipulated in this study was dashboard complexity. The layout or the way the dashboard is presented to the participant defines dashboard complexity. For the purposes of this research, the two dashboard interfaces provided in the survey each had different visual appearance measures that were used to determine which dashboard had less visual complexity as opposed to more visual complexity. Dashboard A has a more clear overall layout, used balanced colors and was less visually distracting; while Dashboard B was more visually dense, there were more visual representations, there was an overuse of poorly-balanced colors, and the overall visual layout was complex or cluttered or dense.

Participant's evaluation of complexity was made by how visually complex or cluttered they were. Four Likert-scale items were used for participant's evaluation of complexity using a standard evaluation method.

The cognitive load was developed as the mediating variable (extraneous cognitive load being defined as the way dashboard information is presented). The amount of effort that a person has to use in their own mind to interpret an approximate amount of information is considered cognitive load. When reviewing the literature on Cognitive Load Theory, it was mentioned that a person may be subjected to excessive extraneous cognitive load when the means by which information is presented to them creates additional effort to interpret that information.

For example in the case of a dashboard, the visual complexity of the dashboard will usually create a greater amount of cognitive load for the individual who must promote the location of relevant information as well as understand the relationship between different metrics.

The dependent variable for this study was viewed by the participants' as perceived decision quality. Perceived decision quality relates to how confident a participant feels regarding his/her ability to make a decision based upon the information being depicted within the dashboard. The intention for organizations to create dashboards is that they can use them as a means to support their decision making process by presenting important information in a form that is easy to read and comprehend.

In the case of dashboards that have a significant amount of cognitive load, the user is much less confident in their ability to interpret the dashboards and make a judgment based on the information available to them. I collected the data on perceived decision quality based upon participant's reported confidence when making a business decision related to the dashboard via four Likert-scale items.

All of the variables measured in the research were done so using five-point Likert scales for each of the participants' responses to statements regarding dashboard complexity, mental effort, and confidence in making decisions after viewing the dashboards. As such, the use of Likert scale measures afforded me the ability to quantify the participants' perceptions of each dashboard and statistically assess the relationships between the variables.

Once the data for the two dashboards (A and B) were collected, I calculated averages of the participant-level construct scores for each of the three constructs for both dashboards. This allowed for a direct comparison of the two conditions and supported the repeated-measures design analysis done as part of the research.

By operationalizing the variables as previously described, I was able to conduct an empirical examination of the relationships proposed in their research model and determined that there was an association between increased loads of cognitive effort and increased complexity of the dashboards as well as that there was a relationship between higher loads of cognitive effort leading to decreased perceived quality of decision-making.

3.5 Data Collection Procedure

The data for this study came from a Google Sheets survey which was distributed to participants via various digital platforms. The survey was administered using the online survey tool Google Forms where respondents could have access and complete the questionnaire from different locations.

The first section of the questionnaire included an introduction to the study and informed participants of how their responses would be used for academic purposes. Informed consent was required for respondents to continue with the survey, which was also mandatory for all participants. The introduction to the survey also mentioned that participation was voluntary and that participants were welcome to terminate the survey whenever they wished.

A few general demographic questions were being collected including an age group, level of education, familiarity with business dashboards and tools for visualizing data, etc., to provide contextual information regarding the sample group while maintaining a relatively concise and simple questionnaire.

After providing consent to participate, all participants went ahead to respond to the rest of the questionnaire. In the next section, participants first viewed screen shots of Business Dashboard A and imagined they were managers examining business performance information related to that dashboard. After being exposed to the dashboard, participants responded to a number of Likert-type questions measuring their perception of dashboard complexity, cognitive load and how well the quality of their decision was made.

Once the participants had completed their assessment of the first dashboard, they were provided a set of screen shots from Business Dashboard B that differed in the level of visual complexity and the overall design quality compared to Business Dashboard A. After viewing the second dashboard, the participants were again requested to respond to the same set of questions as before; thus, providing the ability of the research to evaluate participants' responses for both dashboards using a repeated-measures design. Repeated measures design refers to measuring the same participants under different conditions multiple times.

Completion of the complete questionnaire took the participants approximately five minutes. Upon completion of the data collection period, all responses were exported for statistical analysis.

3.6 Data Analysis

The purpose of the quantitative statistical analysis of the survey data was to explore the relationship between dashboard complexity, cognitive load, and perceived decision quality. Descriptive statistics of all variables in the study were first calculated. Survey items were grouped according to each of the constructs measured: dashboard complexity, cognitive load, and perceived decision quality. The composite scores representing each variable were based on the average of each participant's responses for all survey items of each construct. Means and standard deviations were the primary descriptive statistics used to describe how the participants evaluated the

dashboards, and to identify in general how all participants responded to the dashboards by creating a product based on the total responses of each participant for each construct.

Comparative analysis was then performed to determine whether there were differences between the participants' responses to the two dashboard conditions. As all participants evaluated both dashboard conditions, each participant's responses could be directly compared to determine whether the dashboard with more visual complexity resulted in higher cognitive load and lower perceived quality of decision-making than did the dashboard with less visual complexity. Averages were calculated across all three study variables for both A and B dashboards; the three study variables being the three constructs measured: dashboard complexity, cognitive load, and perceived decision quality. This analysis aided in supporting the repeated measures design of the study, and allowed the two dashboards to be directly compared to each other.

The additional statistical analyses performed on the dashboards included Pearson correlation analyses to explore the relationships among the key variables presented in the theoretical model. Specifically, these analyses examined the relationships between perceived dashboard complexity and cognitive load (i.e., is there a positive correlation between the two) and the relationship between cognitive load and perceived decision quality (i.e., is there a negative correlation between the two).

Statistical analyses were conducted with data analysis tools, including Microsoft Excel and Google Sheets, based on the statistical methods chosen. The tools were used to determine descriptive statistics and Pearson correlation coefficients, thus allowing for the evaluation of the proposed relationships in the research hypothesis.

The results of the study provided empirical evidence regarding the psychological impact of dashboard design on business users by determining whether increased complexity of dashboards resulted in increased cognitive load and whether increased cognitive load impacted how business users evaluated their decision-making ability.

3.7 Ethical Considerations

The research followed all ethical guidelines established to ensure that participant rights and privacy were maintained throughout the research process, including that participant participation was 100% voluntary. As mentioned earlier, each participant received a welcome page that

explained the purpose of the research prior to answering the survey questions. In addition, they were informed that they were to receive informed consent and could end their surveys anytime without penalty.

Finally, the surveys did not gather any personal identifiers, such as Names, Addresses, & Addresses. All survey responses were collected through Google Docs and, therefore, were completely anonymous. All survey responses were used only for research purposes associated with this Thesis Project.

All data collected from the participants in this study were stored securely and accessed only by the Researcher. The data were used solely to conduct the statistical analysis related to the Research Objectives and no responses from individual participants were reported in such a way as to identify which participant's response was included in that report or made public.

The study involved no questions that were sensitive or potentially harmful. Participants only needed to respond to questions in a business dashboard and indicate what their mental effort and decision confidence were; therefore, the risk of participating in this study was low.

The research adheres to established ethical guidelines when conducting studies of humans by providing volunteers with full information about the nature of the study, ensuring their anonymity, and providing for the responsible handling of data.

4. Results and Findings

4.1 Descriptive Statistics

In the end, a total of 101 valid responses were collected and analyzed. All participants evaluated two dashboard interfaces: Dashboard A, designed according to commonly recommended dashboard design principles, and Dashboard B, intentionally designed with visually complex and cluttered features. Participants completed the same questionnaire after viewing each dashboard to compare perceptions across both dashboard conditions.

Descriptive statistics were calculated for dashboard complexity, cognitive load, and perceived decision quality for both dashboards. The results are presented in Table 1

Table 1
Descriptive Statistics for Dashboard Variables

Variable	Mean	Standard Deviation
Dashboard A – Complexity	2.29	0.79
Dashboard A – Cognitive Load	2.46	0.85
Dashboard A – Decision Quality	3.91	0.81
Dashboard B – Complexity	4.46	0.52
Dashboard B – Cognitive Load	4.30	0.63
Dashboard B – Decision Quality	1.86	0.85

There are significant differences in the two types of dashboards shown by the Descriptive Statistics results. Dashboard A received much lower scores in ratings for perceived complexity and

cognitive load, but perceived quality of decisions was rated relatively high. Conversely, Dashboard B had very high ratings for complexity and cognitive load in comparison to Dashboard A, but had much lower ratings in terms of perceived quality of decisions.

Based on these results, Dashboard B was significantly harder for the respondents to understand and required more cognitive effort than Dashboard A; additionally, respondents did not feel as confident about making sound business decisions with Dashboard B compared to Dashboard A. These results further support the fact that a dashboard with a great density of visual display and/or too many sources of visual information can hinder a user's cognitive processing and confidence in making decisions. These conclusions are consistent with earlier research findings on dashboards and cognitive load, which underscore the importance of a visually hierarchical, simple, and clear presentation of information when designing dashboards.

4.2 Correlation Analysis

Pearson correlation analysis was conducted separately for Dashboard A and Dashboard B in order to examine the relationships between dashboard complexity, cognitive load, and perceived decision quality.

Table C2

Dashboard A Correlations

Variables	Correlation (r)
Complexity ↔ Cognitive Load	0.758
Cognitive Load ↔ Decision Quality	-0.574
Complexity ↔ Decision Quality	-0.496

The results for Dashboard A demonstrate a strong positive relationship between perceived dashboard complexity and cognitive load ($r = 0.758$). This indicates that participants who perceived the dashboard as more visually complex also reported higher levels of mental effort when interpreting the dashboard.

A moderate to strong negative relationship was found between cognitive load and perceived decision quality ($r = -0.574$). Higher levels of cognitive load were associated with lower confidence in decision making and reduced perceptions of dashboard usefulness.

Additionally, dashboard complexity demonstrated a moderate negative relationship with perceived decision quality ($r = -0.496$), suggesting that increased complexity may reduce users' confidence in the decisions supported by the dashboard.

Table C3

Dashboard B Correlations

Variables	Correlation (r)
Complexity ↔ Cognitive Load	0.652
Cognitive Load ↔ Decision Quality	-0.629
Complexity ↔ Decision Quality	-0.403

The results for Dashboard B reveal a similarly strong positive relationship between dashboard complexity and cognitive load ($r = 0.652$). Participants who perceived Dashboard B as more visually overwhelming also experienced greater mental effort while interpreting the information presented.

A strong negative relationship was observed between cognitive load and perceived decision quality ($r = -0.629$). Participants experiencing higher cognitive load reported lower confidence in their ability to make reliable business judgments using the dashboard.

Dashboard complexity also demonstrated a moderate negative relationship with perceived decision quality ($r = -0.403$), indicating that dashboards perceived as more visually complex were associated with lower perceived effectiveness for decision support.

4.3 Hypothesis Evaluation

The findings generally support all proposed hypotheses.

H1: Higher levels of dashboard complexity are positively associated with higher levels of users' cognitive load.

The results strongly support H1. For both dashboard conditions, dashboard complexity demonstrated strong positive correlations with cognitive load (Dashboard A: $r = 0.758$; Dashboard B: $r = 0.652$). Participants who perceived dashboards as more visually complex also reported greater mental effort during interpretation.

H2: Cognitive load is negatively associated with perceived decision quality.

The results support H2. Cognitive load demonstrated moderate to strong negative correlations with perceived decision quality in both dashboard conditions (Dashboard A: $r = -0.574$; Dashboard B: $r = -0.629$). Increased mental effort was associated with lower confidence in decision making.

H3: Cognitive load mediates the relationship between dashboard complexity and perceived decision quality.

The findings provide indirect support for H3. Dashboard complexity was negatively associated with perceived decision quality, while cognitive load demonstrated strong relationships with both variables. This pattern suggests that cognitive load may function as an explanatory mechanism linking dashboard complexity to reduced perceived decision quality.

5. Discussion

5.1 Interpretation of Findings

This study indicates that participants respond very differently to the two dashboards. These differences align with what Cognitive Load Theory would predict; more visual complexity means more mental effort gets exerted which results in lower levels of confidence for making a decision.

Dashboard B (The more visually dense/cluttered interface) was rated significantly higher for both cognitive load and complexity when compared to Dashboard A; in contrast, there was a significant decrease in perceived decision quality using Dashboard B versus Dashboard A. The most compelling aspect of both dashboards was the noticeable size of the differences. There was clearly an increased amount of struggle when using Dashboard B, and an associated decrease in confidence when making a business decision based on what they viewed on Dashboard B versus Dashboard A (the more cleanly laid-out/dashboard).

The findings from this study provide substantial support for the theory that presentation of information is just as important to the user's ability to make a decision as is the content presented.

The correlation results added further support. Dashboard complexity and cognitive load were strongly positively correlated in both conditions, meaning participants who found a dashboard visually overwhelming also tended to report higher mental effort — consistently, not just occasionally. Cognitive load in turn showed moderate to strong negative correlations with perceived decision quality, suggesting that as mental effort increased, decision confidence declined.

The pattern across all three variables also points toward cognitive load playing a connecting role between complexity and decision quality, which aligns with the conceptual model developed in the literature review. Dashboard complexity alone showed a weaker relationship with decision quality than cognitive load did, which is exactly what you would expect if cognitive load is part of the explanation for why complex dashboards reduce decision confidence rather than complexity directly causing it.

5.2 Connection to Prior Literature

The conclusion parallels the conclusion of prior empirical research on the significance of simplicity, visual clarity, and cognitive efficiency in Business Intelligence; however, they are based on independent studies. While the past studies of dashboards have routinely suggested that users can have difficulty interpreting information adequately if there is too much data or if the data representations are poorly visualized (Few, 2016; Brath & Peters, 2004), this recent study provides the empirical evidence to support those arguments through demonstrating that users of visually complex dashboards report significantly greater amounts of cognitive load than users of less complex dashboards and report decreased perceived quality of their decisions made when viewing visually complex dashboards.

The relationship found between the complexity of dashboards and cognitive load supports Cognitive Load Theory (Sweller et al., 2011), which posits that an increased representation or disorganized presentation of material causes affect the amount of extraneous cognitive load, through the introduction of additional extraneous demands placed on working memory. In dashboards, the user will be required to process multiple visual components and multiple forms of numerical indicators and labels along with various patterns using a limited amount of cognitive resources available. If a dashboard user's viewing experience is impacted by having visual components that are either visually elaborate or disorganized, the user will likely exhaust their substantial cognitive resources available for processing the data contained in the dashboard.

The present study also supports Burnay et al. (2024), including the findings that the design characteristics of dashboards will create different levels of cognitive and representational load and affect the user's overall experience with the dashboard. Similar to Burnay et al. (2024), the present study found that participants who used a dashboard with too many visual features tried to interpret the dashboard data would report higher cognitive and representational loads than users who used a dashboard that did not have as many visual elements.

The results presented also support the findings of Hjelle et al. (2024), who found that decision-related outcomes on dashboards are significantly influenced by how complex individuals perceive the tasks they perform through the dashboard. While there has been much prior research around dashboard acceptance, usability, and/or system adoption, this study adds to the body of

literature by examining how perceived decision quality is a construct that is directly influenced by the cognitive impact of using a dashboard.

The negative relationship found between cognitive load and perceived decision quality is further supported by the general body of cognitive and decision-making literature. Specifically, Tversky and Kahneman (1974) provided evidence that individuals are constrained by limited cognitive resources, and that when their attention is drawn away from processing information for an extended period of time, the quality of their decisions decreases. Likewise, research on information overload suggests that simply increasing the amount of information available to an individual (i.e., the number of data points displayed on a dashboard) does not necessarily improve the quality of the ensuing decision/and may, in fact, impair their ability to effectively evaluate and integrate that information (Eppler & Mengis 2004; Bawden & Robinson 2009).

The current study further substantiates the body of research that indicates that the effectiveness of dashboards is not only mediated by the technical quality/functionality (or amount of evidence displayed) of the dashboard. Therefore, effective dashboard implementation also relies on alignment between the dashboard design and the user's cognitive abilities.

5.3 Practical Implications

The study findings have several real-world implications for organizations and organizations developing dashboards. Dashboard use is becoming a major component of data-driven decision making; therefore, how the visual design of dashboards impacts cognitive processing and interpretation of data needs to be examined by organizations.

First, this research supports the idea that reducing unnecessary visual complexity can help users process information in the dashboard better. Users may experience greater cognitive burden when dashboards contain too many different colors, competing visual elements, or cluttered layouts, which may make them more likely to lose confidence in making decisions based upon the information presented to them. Therefore, in developing dashboards designed to assist managers or operate the organization, the organization should place a high priority on simplicity, clarity, and visual hierarchy.

Second, this research supports the need for user-centered design in the development of dashboards. Dashboards are often designed by people who are technical experts or analysts and

may have less understanding of cognitive load experienced by non-technical or inexperienced users of the dashboard. This study found that novice or non-expert users of dashboards may be especially cognitively burdened by too much visual overload or complexity of information found on a dashboard. As a consequence, designers of dashboards must ensure that the dashboard is designed in a manner that is consistent with the cognitive ability and level of experience of the user of the dashboard.

Third, organizations should also understand that the functionality of a dashboard can be dependent on the amount of information presented. Dashboards are typically designed to summarize information from many sources into actionable information; however, too much information density may cause confusion instead of helping to interpret data. By strategically using visual grouping, prioritizing information and simplifying layouts, unnecessary contribution of cognitive effort can be reduced while increasing perceived usability and decision confidence.

Finally, the findings indicate that dashboards need to be evaluated based on psychological and cognitive considerations in addition to the technical performance metrics used to evaluate their effectiveness. By evaluating only functionality or quantity of information in dashboards, potentially important aspects of user-centered factors will be missed regarding their ability to interpret and make quality decisions with the information provided.

5.4 Theoretical Implications

By grounding the connection between dashboard complexity and users' perceived quality of decision-making in cognitive load theory, the study extends the growing body of research examining how effective dashboards are. While many prior studies referencing cognitive load theory are discussed in terms of design recommendations and usability, few former studies consider cognitive load as a primary variable in dashboard evaluation research.

Furthermore, the study also helps to advance the existing body of literature on business dashboards by extending analysis beyond adoption-based theories to develop a more psychologically and cognitively-based understanding of effective dashboards. Lastly, based on these findings, it can be posited that dashboard complexity indirectly affects decision-related elements via how users psychologically experience using the dashboard, instead of strictly through usability or system acceptance perceptions.

In addition, the study underscores that the focus of dashboard evaluations should not solely be based upon technical performance, but must also consider human-centered concepts, such as attention, mental effort, and cognitive processing in the evaluation of dashboards, to ensure validity when interpreting results from the dashboard.

As such, the study supports further incorporation of cognitive theory into business intelligence and dashboard research while providing empirical evidence that visually complex dashboards have an adverse effect on the confidence users have in making decisions as a result of increased cognitive load.

6. Limitations and Future Research

6.1 Limitations

Despite contributing to our understanding of the connection between dashboard complexity, cognitive load, and decision quality, the results of this study come with limitations that should be taken into account.

First, the study was undertaken with students and participants early in their careers because these groups were appropriate for examining novice users' abilities to interpret dashboards. This research cannot be generalized to expert managers or data analysts because they have different schemas for thinking about data than beginners do and they have a lot more experience working with complex dashboards; therefore, this could reduce the cognitive demands experienced by them.

Second, participants were asked to think about how they perceived their cognitive load and decision quality when they were interpreting dashboard data. The use of self-reported surveys is an accepted method of collecting subjective data in cognitive load and usability studies. Self-report surveys cannot necessarily provide an accurate and objective account of the amount of cognitive processing or the performance of an objective decision because user responses can be affected by individual interpretations, attention differences, and prior experiences with dashboards.

Another limitation is this study employed static screenshots as dashboards as opposed to fully functioning, interactive dashboard systems. With dashboards in professional organizations, users interact dynamically with, through filtering, drilling down into, and changing the way they see the data. The lack of interactivity may decrease the study's ecological validity and the ability to generalize its results to actual dashboard use in organizations.

Additionally, the study examined visual complexity and cognitive load only in a relatively controlled laboratory environment. It did not explore additional variables that could affect a user's evaluation of the quality of a decision made from dashboard data, such as a user's time pressure, degree of familiarity with the metrics on a dashboard, and domain knowledge. These variables have the potential to affect how users process data presented on dashboards, and subsequently, evaluate the quality of their decision-making.

The findings demonstrated that dashboard complexity, cognitive load, and perceived decision quality were significantly related; however, the study was designed as not experimental but correlational and cross-sectional. There are no definitive conclusions regarding an established causal relationship. Although the results are consistent with Cognitive Load Theory and previous research, there remain additional issues including the need for further experimental and longitudinal studies to provide fuller understanding of these behaviours over time.

6.2 Future Research Directions

The findings of this study present numerous avenues for future research. For instance, additional research can be conducted using experienced business professionals in other words, managers or analysts, in an organizational setting and compare novice users with expert users so researchers can understand if professional experience acts as a moderating variable in the relationship between dashboard complexity and cognitive load.

Future studies may also examine the effects of adding objective performance-based metrics such as task accuracy, response time, effectiveness of decisions to the subjective perceptions of participants while they are interacting with a dashboard. By assessing objective measures of performance in addition to subjective measures, researchers can determine if increased cognitive load has a direct relationship with objective outcomes of decision making and the perceived level of confidence in the decisions made.

Another avenue for future research could be looking at interactive dashboards compared to using static screenshots of dashboards. There are certain features like filtering, drill down capability, adaptable visualization, and user-customized layouts of interactive dashboards that may influence cognitive load in a different manner than static dashboards. As more companies begin utilizing real-time and interactive business intelligence systems, it will be important to understand the relationships between cognitive load and different dashboard designs.

Additionally, future studies may incorporate physiological or behavioral measurements of cognitive load (e.g., eye tracking, gaze tracking, and attention monitoring technologies) to explain how attention and visual focus may affect how participants interpret dashboards and process the information contained within the dashboard.

Future research has the potential to evaluate and develop new dashboard design factors beyond visual complexity, such as color harmony, storytelling, animation, personalisation, mobile responsiveness and artificial intelligence (AI) to determine how these may alter users' cognitive processing, trust and experience when making decisions.

Additionally, future research could evaluate the effectiveness of dashboards in different industries and decision-making contexts outside of visual static context, including healthcare, finance, supply chain and public policy. Requirements and cognitive load associated with the use of dashboards could differ greatly depending on the goals of the organisation, the level of experience of users with dashboards and the urgency of making decisions.

In summary, the future of dashboard research should validate and incorporate cognitive and psychological frameworks when evaluating dashboards to allow researchers to gain insights into how dashboard design will impact users' understanding, focus and decision-making.

7. Conclusion and Recommendations

7.1 Conclusion

The use of business dashboards is increasing in popularity as a means for organizations to communicate their performance and to support their decision-making. However, despite business dashboards' widespread use, there is a lack of empirical research focused on the cognitive experience of using business dashboards. The purpose of this study was to address this gap by investigating whether the complexity of a business dashboard increases users' cognitive load (mental effort), which in turn would decrease users' confidence in making decisions.

The findings of this study were unequivocal. Participants reported that they had significantly greater difficulty understanding the visually complex or "dense" business dashboard, that they used significantly greater mental effort in working with the business dashboard, and that they had significantly less confidence in making sound business decisions based on the business dashboard. The opposite pattern of results was found for the visually simple or "sparse" business dashboard. The difference in user experience between the two types of dashboards was substantial enough to warrant attention beyond an academic context.

Correlation analyses provided additional insight into the results obtained in this study. Dashboard complexity and cognitive load were found to be highly correlated, and cognitive load also demonstrated a significant negative relationship with perceived decision quality. The results of the present study indicate that users of business dashboards have a consistent and similar response to visually complex or "overloaded" business dashboards, which is not attributable to the design of either business dashboard.

What this ultimately points to is something practitioners often overlook: adding more information to a dashboard does not make it more useful. If anything, the opposite may be true when that information is poorly organized or visually competing for attention. Cognitive Load Theory helps explain why this is the case. Working memory has limits, and when a dashboard pushes against those limits, users' ability to think clearly about the data suffers.

There are genuine limitations here, particularly around the sample size and the use of static dashboard images rather than live interactive systems, and these should temper how broadly the findings are applied. But as an exploratory study, the results make a coherent case that dashboard design is not just an aesthetic concern, it is a cognitive one, with real consequences for how confidently people use data to make decisions.

7.2 Recommendations

Several recommendations for organizations and dashboard designers are present in the data collected from this research.

First, organizations need to reduce the overall number of visual components within dashboards designed for helping make business decisions through increasing visually clear and simple dashboards containing minimal visual information with no distractions will not only alleviate some cognitive burden placed on the user when interpreting insights obtained via dashboards, but will give the user greater confidence in his/her ability to correctly interpret all underlying business insights based upon the information included within the dashboard.

When designing dashboards, designers should consider users' cognitive limits and visual attention limits through user-centered design principles, including the utilisation of visually organised and structured representations of information in combination with clarity, including the use of colors that establish a visual hierarchy (as supported by Wu et al.; 2022) to allocate attention to particular aspects of the information contained within a dashboard. The research findings indicate that the presentation of large amounts of data does not inherently enhance decision making for a user (e.g., improving a user's ability to process the data effectively), but that excessive information and complex data structures could impair users' ability to process data effectively. Therefore, when designing dashboards, organizations should give priority to the relevance and usefulness of the data presented over providing as many pieces of data as possible.

An evaluation should include both a dashboard's technical capability and a dashboard's cognitive impact upon the user. Usability testing, cognitive load evaluation, and user input on how to utilize a dashboard's capabilities should all be incorporated during the design of a dashboard to increase its effectiveness for all users within a given organization.

Lastly, user training and onboarding are an area for improvement to assist users with interpreting an often-complex dashboard system, especially where significant amounts of data require a user to process them completed most efficiently. Helping users develop a better cognitive understanding of the way a dashboard's data is structured and visualized will ultimately lessen the degree of mental effort they expend when interpreting the data within the dashboards and build greater confidence in their ability to make informed decisions.

In summary, this study demonstrates the necessity of developing dashboards to allow for efficient cognitive processing and effective interpretations within an organization that has increasingly transitioned to a data-driven environment.

Bibliography

Bawden, D., & Robinson, L. (2009). The dark side of information: Overload, anxiety and other paradoxes and pathologies. *Journal of Information Science*, 35(2), 180–191. <https://doi.org/10.1177/01655551508095781>

Bera, P. (2016). How colors in business dashboards affect users' decision making. *Communications of the ACM*, 59(4), 50–57. <https://doi.org/10.1145/2818993>

Brath, R., & Peters, M. (2004). Published in DM Review Online – October 15, 2004 Printed from DMReview.com.

Burnay, C., Bouraga, S., & Lega, M. (2023). When Dashboard's Content Becomes a Barrier—Exploring the Effects of Cognitive Overloads on BI Adoption. In S. Nurcan, A. L. Opdahl, H. Mouratidis, & A. Tsohou (Eds.), *Research Challenges in Information Science: Information Science and the Connected World* (Vol. 476, pp. 435–451). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-33080-3_26

Burnay, C., Lega, M., & Bouraga, S. (2024). Business intelligence and cognitive loads: Proposition of a dashboard adoption model. *Data & Knowledge Engineering*, 152, 102310. <https://doi.org/10.1016/j.datak.2024.102310>

DELONE, W., & MCLEAN, E. (2003). The DeLone and McLean Model of Information Systems Success: A Ten-Year Update. *Journal of Management Information Systems*, 19(4), 9–30. <https://doi.org/10.1080/07421222.2003.11045748>

Emami, Z., & Chau, T. (2020). The effects of visual distractors on cognitive load in a motor imagery brain-computer interface. *Behavioural Brain Research*, 378, 112240. <https://doi.org/10.1016/j.bbr.2019.112240>

Eppler, M. J., & Mengis, J. (2004). The Concept of Information Overload: A Review of Literature from Organization Science, Accounting, Marketing, MIS, and Related Disciplines. *The Information Society*, 20(5), 325–344. <https://doi.org/10.1080/01972240490507974>

Few, S. (2016). Data-Visualization-for-Human-Perception. <https://journalismcourses.org/wp-content/uploads/2020/07/Data-Visualization-for-Human-Perception.pdf>

Hjelle, S., Mikalef, P., Altwaijry, N., & Parida, V. (2024). Organizational decision making and analytics: An experimental study on dashboard visualizations. *Information & Management*, 61(6), 104011. <https://doi.org/10.1016/j.im.2024.104011>

Kelly, N. (n.d.). Delivering Data Analytics. Retrieved <https://deliveringdataanalytics.com/i-tried-my-hand-at-designing-the-worst-dashboard-i-could-imagine/>

Maedche, A. [Hrsg, Brocke, J. V. [Hrsg, & Hevner, A. [Hrsg. (2017). Designing the Digital Transformation: DESRIST 2017 Research in Progress Proceedings [PDF]. <https://doi.org/10.5445/IR/1000069452>

Nontokozo, C. M., & Mkhwanazi, S. (2025, October). Designing Digital Dashboards for Business Performance: Principles and Evidence. <https://doi.org/10.21203/rs.3.rs-7817897/v1>

Orlovskyi, D., & Kopp, A. (n.d.). A Business Intelligence Dashboard Design Approach to Improve Data Analytics and Decision Making.

Rieg, R. (2025). Exploring the determinants and performance effects of digital dashboard use by management accountants. *Journal of Management Control*. <https://doi.org/10.1007/s00187-025-00400-0>

Rosenholtz, R., Li, Y., & Nakano, L. (2007). Measuring visual clutter. *Journal of Vision*, 7(2), 17. <https://doi.org/10.1167/7.2.17>

Shao, H., Martinez-Maldonado, R., Echeverria, V., Yan, L., & Gasevic, D. (2024). Data Storytelling in Data Visualisation: Does it Enhance the Efficiency and Effectiveness of Information Retrieval and Insights Comprehension? Proceedings of the CHI Conference on Human Factors in Computing Systems, 1–21. <https://doi.org/10.1145/3613904.3643022>

Sweller, J. (1988). Cognitive Load During Problem Solving: Effects on Learning. *Cognitive Science*, 12(2), 257–285. https://doi.org/10.1207/s15516709cog1202_4

Tversky, A. (1974). JUDGMENT UNDER UNCERTAINTY: HEURISTICS AND BIASES.

Wu, Q., Tan, Z., & Liu, J. (2022). Experimental study on the relationship between the harmony and cognitive load of business intelligence dashboard color combinations. *Color Research & Application*, 47(4), 920–941. <https://doi.org/10.1002/col.22768>

Xie, J., Xu, G., Luo, A., Li, M., Zhang, S., Han, C., & Yan, W. (2017). The Role of Visual Noise in Influencing Mental Load and Fatigue in a Steady-State Motion Visual Evoked Potential-Based Brain-Computer Interface. *Sensors*, 17(8), 1873. <https://doi.org/10.3390/s17081873>

Yigitbasioglu, O. M., & Velcu, O. (2012). A review of dashboards in performance management: Implications for design and research. *International Journal of Accounting Information Systems*, 13(1), 41–59. <https://doi.org/10.1016/j.accinf.2011.08.002>

Appendix A

Survey Questionnaire

The survey questionnaire used in this study was designed to measure participants' perceptions of dashboard complexity, cognitive load, and perceived decision quality after viewing two business dashboard interfaces. Survey items were adapted from prior research on information overload and visual clutter (Eppler & Mengis, 2004; Rosenholtz, 2007), Cognitive Load Theory (Sweller, 1988), and business intelligence dashboard effectiveness research (Burnay et al., 2024; Hjelle et al., 2024).

Participants viewed two dashboard interfaces and completed the same questionnaire after viewing each dashboard. This repeated-measures design allowed comparisons between participants' perceptions across the two dashboard conditions.

Participants responded to each statement using a five-point Likert scale:

Scale Value	Meaning
1	Strongly disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly agree

Section A: Dashboard Complexity

1. The dashboard appears visually complex.
2. The use of colors in the dashboard makes it harder to focus on key information.
3. The layout of the dashboard feels cluttered.

4. The dashboard contains many visual elements that compete for attention.

Section B: Cognitive Load

5. Interpreting the information on this dashboard required mental effort.
6. I had to concentrate carefully to understand the information presented.
7. The dashboard required significant effort to interpret.
8. I could easily understand what the dashboard was showing. (reverse-coded)

Section C: Perceived Decision Quality

9. I would feel confident making a decision based on the information presented in this dashboard.
10. The dashboard provides clear information to support decision making.
11. I believe I could make a reliable business judgment using this dashboard.
12. The information presented in the dashboard is sufficient for making decisions.

Participants completed the questionnaire twice: once after viewing Dashboard A and once after viewing Dashboard B. Dashboard A represented a more visually organized and simplified dashboard design, while Dashboard B represented a more visually dense and complex dashboard condition.

The questionnaire items were adapted from validated academic scales to ensure content validity. Specifically, constructs for load were derived from Burnay et al. (2024), decision quality from Rieg (2025), and information quality dimensions from Hjelle et al. (2024)

Appendix B

Dashboard Stimuli

Two dashboard interfaces were used as visual stimuli in this study. Participants viewed both dashboards and completed the same questionnaire after viewing each interface.

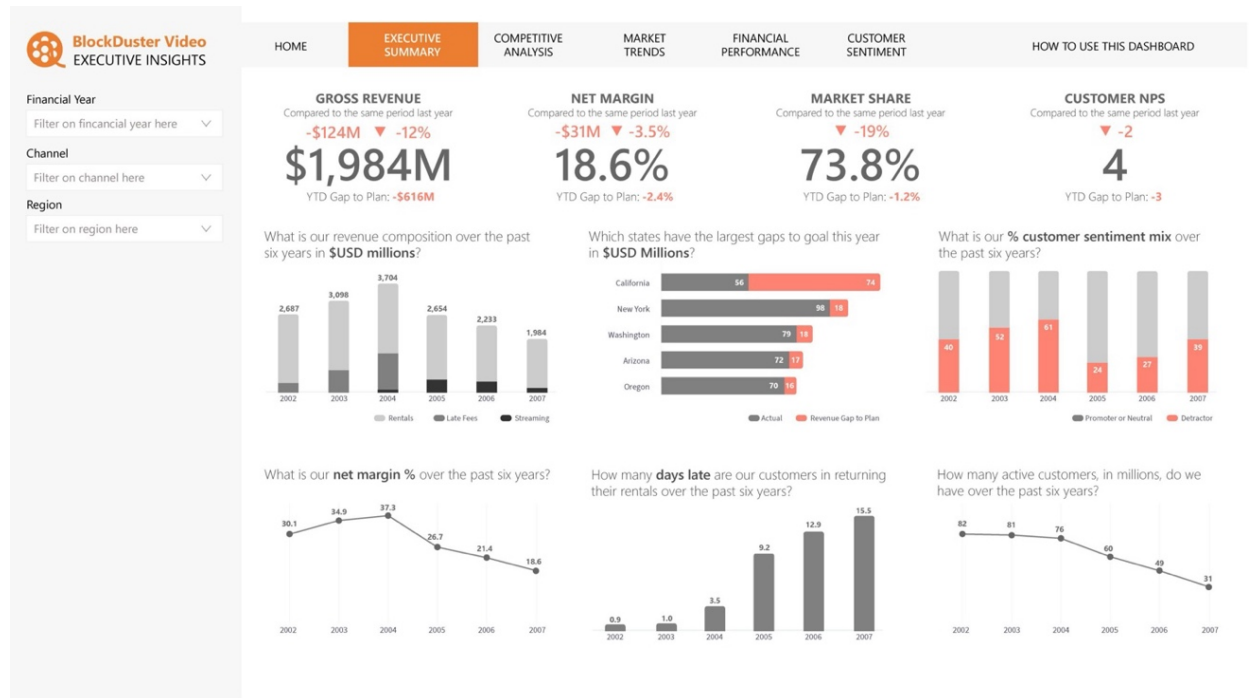
Dashboard A represented a higher-quality dashboard condition designed according to commonly recommended dashboard design principles, including clearer organization, stronger visual hierarchy, reduced clutter, and more balanced use of colors and visual elements.

Dashboard B represented a lower-quality dashboard condition characterized by excessive visual density, competing visual elements, cluttered layout structure, and visually overwhelming design characteristics.

Both dashboard images were adapted from publicly available dashboard examples published by Delivering Data Analytics (n.d.) and were used solely for academic research purposes to evaluate participants' perceptions of dashboard complexity, cognitive load, and perceived decision quality.

Figure B1

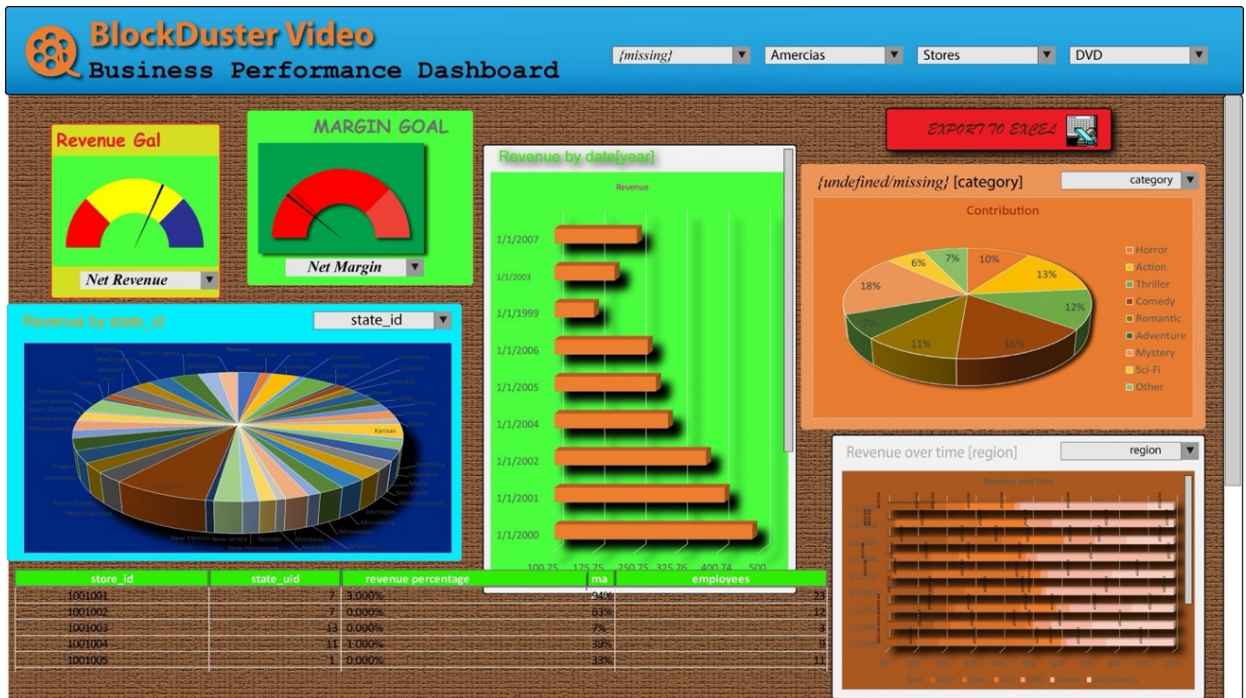
Dashboard A – Higher-Quality Dashboard Condition



Source: Adapted from Delivering Data Analytics (Kelly, n.d.)

Figure B2

Dashboard B – Lower-Quality Dashboard Condition



Source: Adapted from Delivering Data Analytics (Kelly, n.d.)

Appendix C

Descriptive Statistics and Correlation Tables

Table C1

Descriptive Statistics for Dashboard Variables

Variable	Mean	Standard Deviation
Dashboard A – Complexity	2.29	0.79
Dashboard A – Cognitive Load	2.46	0.85
Dashboard A – Decision Quality	3.91	0.81
Dashboard B – Complexity	4.46	0.52
Dashboard B – Cognitive Load	4.30	0.63
Dashboard B – Decision Quality	1.86	0.85

Table C2

Dashboard A Correlations

Variables	Complexity	Cognitive Load	Decision Quality
Complexity	1.000	0.758	-0.496
Cognitive Load	0.758	1.000	-0.574
Decision Quality	-0.496	-0.574	1.000

Table C3

Correlation Analysis for Dashboard B Variables

Variables	Complexity	Cognitive Load	Decision Quality
Complexity	1.000	0.652	-0.403
Cognitive Load	0.652	1.000	-0.629
Decision Quality	-0.403	-0.629	1.000

Table C4

Participant Information

Category	Frequency
Total Valid Responses	101

Note: Participants primarily consisted of students and early-career individuals familiar with dashboard interfaces and business-related visualizations.

Pearson correlation analysis was conducted to examine relationships between dashboard complexity, cognitive load, and perceived decision quality across both dashboard conditions.

Positive correlation values indicate positive relationships between variables, while negative values indicate inverse relationships.

Dashboard A represented the higher-quality dashboard condition designed according to commonly recommended dashboard design principles. Dashboard B represented the lower-quality dashboard condition characterized by greater visual complexity and visual clutter.

Appendix D

Declaration of Generative AI Use

Declaration table by section

Thesis section	AI used (Yes/No)	Type of use (rewording, idea generation, etc.)	Prompts used (verbatim)	Verification methods applied
Issues	Yes	Grammar, counterarguments, feedback on my understanding of issues	“Please check my grammar and spelling for the following write up” “Please give me a counterargument to my text and feedback on the development of my idea”	Cross checked information, reviewed all edits made, verified sources
Literature review	Yes	Grammar, understanding literature better	“Please check my grammar and spelling for the following write up” “Please ask me questions to verify my understanding of the literature” “Please check to see if my understanding seems thorough and clear for the following write up”	Cross checked information, reviewed all edits made, verified sources
Methodology	Yes	Grammar, spelling, checking if my logic holds	“Please check my grammar and spelling” “Please check to see if my logic holds	Cross checked information, reviewed all edits made, verified sources
Analysis of results	No			
Discussion	No			
Conclusions	No			

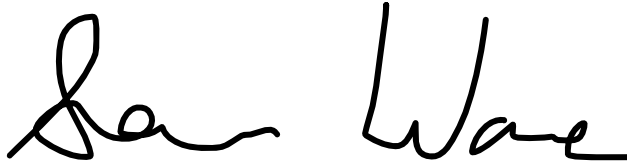
Statutory declaration

"I, the undersigned WARE SAVANNA hereby certify that:

1. This work reflects my personal thinking and critical analysis skills
2. I have declared all use of generative AI in a comprehensive and transparent manner
3. The identification of the gap, the formulation of the problem, the construction of the conceptual framework, the empirical data, the interpretation and critical analysis of the results, the discussion, and the conclusion represent my original intellectual work

4. I understand that any breach of these commitments constitutes
academic fraud

22nd May 2026

A handwritten signature in black ink, appearing to read 'Sam Wane'. The signature is fluid and cursive, with the first name 'Sam' and the last name 'Wane' clearly distinguishable.

This statement constitutes the **Declaration of Generative AI Use** required by **EDC Paris Business School** in accordance with the *Student Charter on the Use of Generative AI in Final Year Projects*. This declaration is submitted to ensure compliance with the principles of academic integrity, transparency, and personal intellectual responsibility established by EDC Paris Business School. All analyses, interpretations, conclusions, and original contributions presented in this thesis remain the result of the author's own critical thinking and academic work.